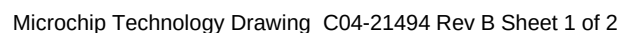
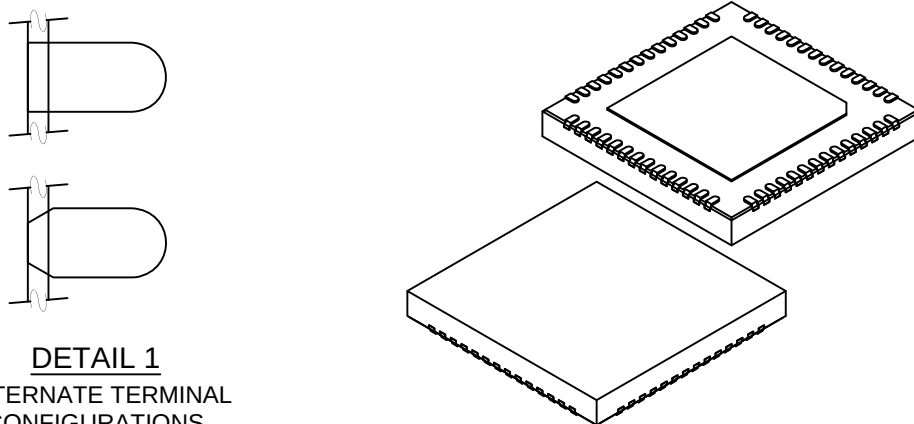


Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



56-Lead Very Thin Plastic Quad Flat, No Lead Package (TYB) - 7x7 mm Body [VQFN] With 4.60 mm Exposed Pad and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



DETAIL 1

ALTERNATE TERMINAL
CONFIGURATIONS

		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		56		
Pitch	e		0.40 BSC		
Overall Height	A		0.80	0.85	0.90
Standoff	A1		0.00	0.02	0.05
Terminal Thickness	A3		0.203 REF		
Overall Length	D		7.00 BSC		
Exposed Pad Length	D2		4.50	4.60	4.70
Overall Width	E		7.00 BSC		
Exposed Pad Width	E2		4.50	4.60	4.70
Optional Index Chamfer	CH		-	0.35	-
Terminal Width	b		0.15	0.20	0.25
Terminal Length	L		0.35	0.40	0.45
Terminal-to-Exposed-Pad	K		0.80 REF		
Wettable Flank Step Length	D3		-	-	0.085
Wettable Flank Step Height	A4		0.10	-	0.19

Dimensions D3 and A4 above apply to all new products released after November 1, and all products shipped after January 1, 2019, and supersede dimensions D3 and A4 below.

No physical changes are being made to any package; this update is to align cosmetic and tolerance variations from existing suppliers.

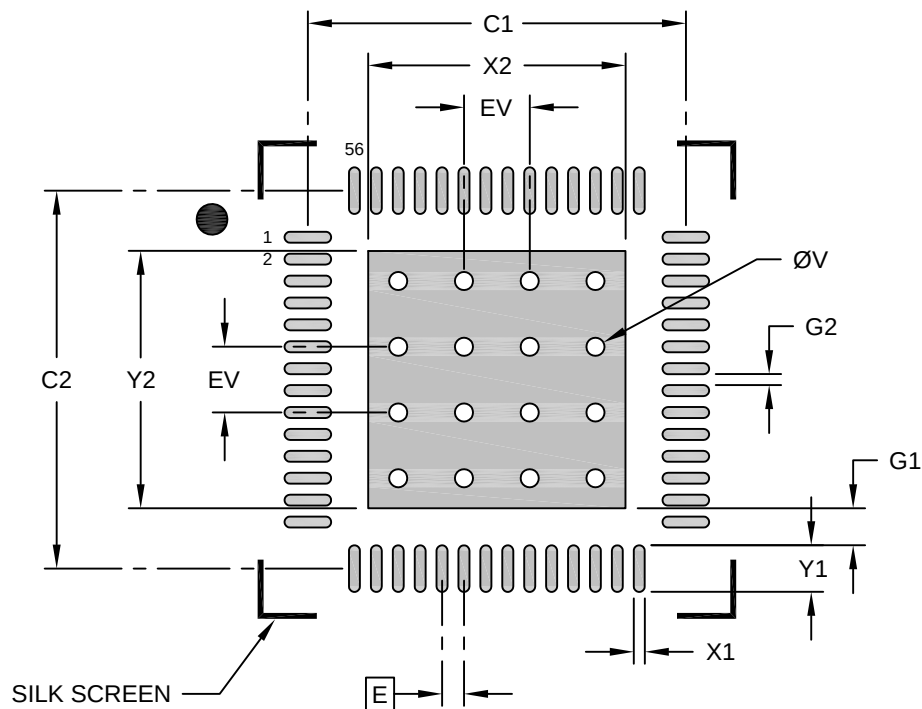
Wettable Flank Step Length	D3	0.035	0.06	0.085
Wettable Flank Step Height	A4	0.10	-	0.19

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

56-Lead Very Thin Plastic Quad Flat, No Lead Package (TYB) - 7x7 mm Body [VQFN] With 4.60 mm Exposed Pad and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E		0.40 BSC	
Optional Center Pad Width	X2			4.70
Optional Center Pad Length	Y2			4.70
Contact Pad Spacing	C1		6.90	
Contact Pad Spacing	C2		6.90	
Contact Pad Width (X56)	X1			0.20
Contact Pad Length (X56)	Y1			0.80
Contact Pad to Center Pad (X56)	G1	0.30		
Contact Pad to Contact Pad (X52)	G2	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process